

Grant Achuzia

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EDUCATION

Carleton University

September 2021 - April 2026

Bachelor of Engineering (B.Eng.), Computer Systems Engineering

Ottawa, ON

- Coursework: Data Structures & Algorithms, Embedded Systems, Operating Systems, Computer Architecture
- Awards: **2024** Canadian Engineering Competition: **Bronze Medalist**, National Programming Category (advanced through university and regional rounds, against **10+** university teams)

EXPERIENCE

Lumentum

January 2025 – August 2025

System Software & Design Verification Intern

Ottawa, ON

- Architected an end-to-end TypeScript test automation suite with CI/CD pipelines via GitHub Actions and Selenium/Tauri E2E, cutting manual regression cycles from **several hours** down to **under 30 minutes** per release
- Built a plugin-based Python CLI tool that auto-parses device logs and renders PPTX/PDF verification reports, eliminating manual report assembly for **12 engineers** across **3 teams**
- Automated high-risk firmware upgrades and connectivity validation across **5+** networked devices using Bash and gNOI/gNOIC, saving an estimated **3–5+** collective engineer-hours weekly on routine maintenance workflows

Maintenance Drone Co.

May 2024 – December 2024

Software Engineering Intern

Ottawa, ON

- Cut 3D point-cloud processing latency by an estimated **15–20%** using NumPy grid downsampling and noise filters, improving pipeline throughput while preserving 3D model accuracy in Blender for production drone surveys
- Engineered a Shapely/Matplotlib path-planning tool that parsed KML/XML into optimized waypoint missions, replacing **3+ hours** of manual route digitization per survey with automated generation across operations

Roof Maintenance Solutions

May 2022 – December 2023

Software Engineering Intern

Ottawa, ON

- Built Python scripts to automate high-volume XML annotation transfers across **100+ survey files** and **1GB+** of data, reducing report generation from roughly **10 minutes** to **under 1 minute** per batch
- Implemented automated Regex validation across the same survey pipeline, routinely flagging **10+** data-quality issues per large batch & eliminating costly re-inspections caused by data corruption

PROJECTS

CU-Bytes

github.com/GAchuzia/cu-bytes

Engineering Capstone | PyTorch, Flask, React Native/Expo, SQLite, Azure

- Trained and deployed a ResNet50-based computer vision model achieving **87% accuracy** across **176 food categories** from photos; optimized via transfer learning and custom data augmentation
- Built a modular training framework supporting architecture hot-swapping (ResNet50/EfficientNet/MobileNet), checkpoint resumption, and automated accuracy regression tests
- Designed and shipped a cross-platform React Native frontend with modern UI components running on iOS, Android, and desktop; presented to **50+** users at capstone fair

Godot Engine

github.com/godotengine/godot-docs

Open Source Contributor

- Submitted fixes to the official contributor documentation: corrected an outdated tooltip API reference (`set_tooltip` to `set_tooltip_text`) deprecated since 4.0, and improved IDE setup cross-linking for new contributors

SKILLS

Languages: Python, TypeScript/JavaScript, C/C++, Bash, SQL

Frameworks & Libraries: React, React Native, Flask, PyTorch, Node.js, NumPy, Shapely, Matplotlib

Tooling & Infrastructure: Git, GitHub Actions, Docker, Linux, Selenium, Jira, Agile/Scrum